MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

Data Science Mini Project Synopsis

Mohit 230932334	Vibhu Kamath 23032152	Nikilesh B 230932158	Prassan 230932154
E&I 6 Sem	E&I 6 Sem	E&I 6 Sem	E&I 6 Sem
Ph.no:8307356509	Ph.no 8484991298	Ph.no:8151876724	Ph.no 6203347213

1.Title

Customer Transaction Analysis, Churn Prediction, and Revenue Optimization using Machine Learning and Data Visualization Techniques.

2.Team Details

Mohit 230932334	Vibhu Kamath 23032152	Nikilesh B 230932158	Prassan 230932154
E&I 6 Sem	E&I 6 Sem	E&I 6 Sem	E&I 6 Sem
Ph.no:8307356509	Ph.no 8484991298	Ph.no:8151876724	Ph.no 6203347213

Data Science Mini Project

Mentor: Mrs Munmun Gharami

Assistant Professor

School of Computer Engineering

2.Abstract

In the contemporary business environment, organizations generate significant volumes of customer transaction data through digital platforms and retail operations. Analysing this data effectively can provide valuable insights into customer purchasing behaviour, revenue patterns, and long-term customer value. However, many organizations lack structured analytical approaches to convert raw transactional data into meaningful business insights. As a result, important patterns related to customer retention, churn behaviour, and revenue contribution often remain unexplored.

This project focuses on the analysis of customer transaction data to understand purchasing patterns and identify factors that influence customer retention and revenue generation. The study applies data science techniques to process and analyse transactional datasets containing information such as customer identifiers, transaction dates, product details, and purchase values. Key analytical metrics including customer lifetime value, purchase frequency, and revenue contribution will be calculated to evaluate customer behaviour.

Machine learning techniques will be used to perform customer segmentation and to develop models that help identify potential customer churn. Additionally, data visualization tools will be utilized to present analytical insights through interactive dashboards. The proposed approach demonstrates how data-driven methods can assist businesses in understanding customer behaviour and improving decision-making related to customer engagement and revenue optimization.

3.Introduction

In the modern digital economy, businesses generate large volumes of transactional data through customer purchases and interactions. This data contains valuable information about customer behaviour, purchasing patterns, and revenue generation. Proper analysis of such data enables organizations to understand their customers more effectively and develop strategies that improve customer retention and business performance.

Customer transaction data can provide insights into how frequently customers make purchases, how much revenue they contribute, and how their behaviour changes over time. However, many organizations are unable to fully utilize this information due to the lack of analytical tools and data-driven frameworks. As a result, important patterns related to customer churn, purchasing behaviour, and revenue trends often remain undiscovered.

Data science techniques and machine learning methods provide an effective approach for analysing large datasets and identifying meaningful patterns. By applying these techniques to customer transaction data, it becomes possible to classify customers based on their purchasing behaviour, predict potential customer churn, and evaluate key metrics such as customer lifetime value and purchase frequency.

This project focuses on analysing customer transaction data to extract insights related to customer behaviour and revenue generation. Machine learning techniques will be applied to perform customer segmentation and churn prediction, while data visualization tools will be used to present analytical results in an understandable format. The study aims to demonstrate how data-driven analysis can support better decision-making and help organizations improve their customer retention and revenue optimization strategies.

4.Literature Review

Customer analytics has gained significant importance in recent years due to the increasing availability of large-scale transactional data generated by businesses. Organizations rely on data analysis techniques to understand customer purchasing behaviour, improve marketing strategies, and enhance revenue generation. Data mining and machine learning approaches have been widely applied to analyse customer data and extract meaningful patterns that support business decision-making [1].

Customer segmentation is one of the most studied areas in customer analytics. Researchers have used clustering algorithms such as K-Means to group customers based on their purchasing behaviour, transaction frequency, and spending habits. Such segmentation allows businesses to identify high-value customers and develop targeted marketing strategies that improve customer engagement and retention [2].

Another important concept discussed in the literature is Customer Lifetime Value (CLV), which measures the total revenue contribution of a customer over a period. Studies suggest that analysing CLV helps organizations allocate marketing resources more effectively and focus on profitable customers [3].

Researchers have also emphasized the importance of Recency, Frequency, and Monetary (RFM) analysis for evaluating customer behaviour. RFM analysis helps categorize customers according to their purchasing recency, purchase frequency, and overall spending, which supports more effective customer segmentation and marketing strategies [4].

Predictive analytics has also been applied to identify potential customer churn. Machine learning models such as logistic regression, decision trees, and random forest algorithms are frequently used to predict whether customers are likely to discontinue purchasing. These models enable businesses to implement retention strategies before customers leave [5].

In addition, several studies highlight the role of data visualization techniques in presenting analytical insights effectively. Visualization tools allow decision-makers to understand complex data patterns through interactive charts, graphs, and dashboards [6].

Recent developments in business intelligence platforms have enabled organizations to integrate analytical results with real-time dashboards for monitoring key performance indicators. Such dashboards help businesses track revenue trends, customer activity, and retention metrics in an efficient manner [7].

Furthermore, research in the field of customer relationship management (CRM) has shown that data-driven decision making improves customer satisfaction and long-term business performance. Analytical models can assist organizations in understanding customer preferences and improving service quality [8].

With the growing adoption of data science in business environments, integrating machine learning models with customer analytics frameworks has become an effective approach for identifying revenue opportunities and reducing customer churn [9].

Overall, the literature indicates that the combination of machine learning techniques, customer behaviour analysis, and interactive data visualization provides a powerful framework for understanding customer transactions and optimizing revenue strategies [10].

5.Problem Statement

In the modern digital economy, organizations generate large volumes of customer transaction data through online platforms and retail systems. This data contains valuable information related to customer purchasing behaviour, spending patterns, and revenue generation. However, many organizations are unable to effectively analyse this data due to the absence of structured analytical frameworks and data-driven tools. As a result, important insights regarding customer behaviour and revenue contribution often remain unidentified [1], [2].

Another major challenge faced by businesses is identifying customers who are likely to discontinue purchasing. Customer churn can significantly affect business revenue and long-term profitability. Without appropriate analytical models, organizations are unable to predict customer churn or identify high-value customers effectively [3], [5].

Therefore, there is a need for a systematic data-driven approach that can analyse customer transaction data and extract meaningful insights. By applying machine learning techniques and

data analytics methods, it becomes possible to perform customer segmentation, predict potential churn, and evaluate key metrics such as customer lifetime value and purchase frequency. Such an analytical framework can assist organizations in improving customer retention and optimizing revenue strategies [2], [9].

6.Objective

The main objectives of this project are:

1. To analyse customer transaction data to understand purchasing behaviour and revenue patterns.
2. To identify key customer analytics metrics such as customer lifetime value, purchase frequency, and retention rate.
3. To perform customer segmentation using machine learning techniques based on purchasing behaviour.
4. To develop a predictive model for identifying potential customer churn.
5. To visualize analytical insights through an interactive dashboard for better understanding of business trends.

7.Methodology

The proposed methodology follows a structured data science pipeline for analysing customer transaction data and extracting meaningful business insights. The process begins with collecting a suitable dataset containing customer transaction information such as customer identifiers, transaction timestamps, product categories, quantities purchased, and revenue values.

In the first stage, data preprocessing is performed to ensure data quality and consistency. This step includes handling missing values, removing duplicate records, converting date-time formats, and performing data normalization where required. After preprocessing, the dataset is prepared for further analytical operations.

The next stage involves Exploratory Data Analysis (EDA), where statistical summaries and visualizations are used to identify patterns in customer transactions, purchase frequency, and revenue distribution. This step helps in understanding the overall characteristics of the dataset and identifying trends or anomalies.

Following EDA, feature engineering is applied to derive important behavioural features from the raw data. These features include Recency (time since last purchase), Frequency (number of purchases), and Monetary value (total spending), commonly referred to as RFM analysis. Additional features such as average order value and purchase intervals may also be derived to enhance the analytical model.

Next, machine learning techniques are applied to analyse customer behaviour. Clustering algorithms such as K-Means are used for customer segmentation based on purchasing patterns, while classification algorithms such as Logistic Regression or Random Forest may be used to predict potential customer churn.

The trained models are then evaluated using appropriate performance metrics. Finally, the analytical results are presented using interactive data visualization dashboards, which display key performance indicators and insights that support data-driven decision-making.

Dataset Parameters Used for Analysis

Parameter	**Description**
Customer_ID	Unique identifier assigned to each customer
Transaction_ID	Unique identifier for each transaction
Transaction_Date	Date on which the purchase was made
Product_Category	Category of product purchased
Quantity	Number of items purchased in the transaction
Unit_Price	Price of a single item
Revenue	Total transaction value (Quantity × Unit Price)
Customer_Location	Geographic region or country of the customer

Machine Learning Model Outputs

Output	**Description**
Customer Segment	Group assigned by clustering model (e.g., high-value, loyal, at-risk)
Churn Probability	Probability that a customer will stop purchasing
Predicted Customer Value	Estimated future revenue contribution
Customer Retention Indicator	Classification of active or churn-risk customers

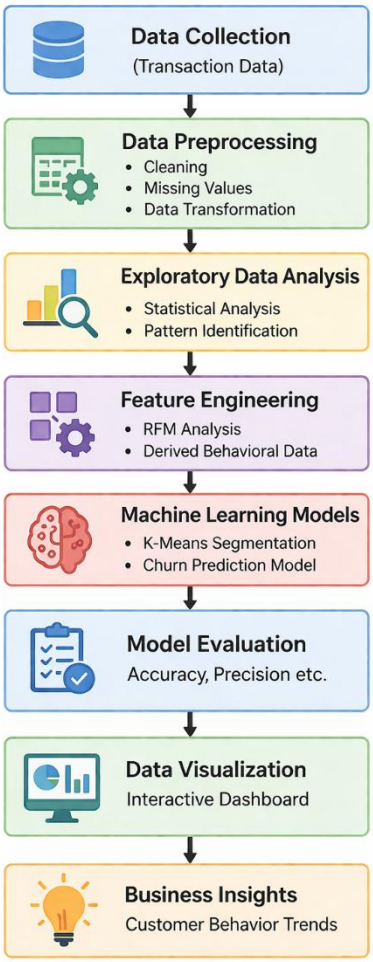


Fig 1: Flow Diagram of the Proposed System

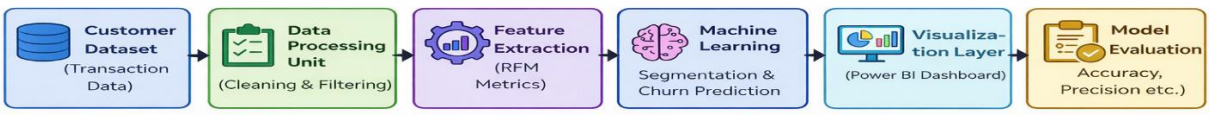


Fig 2: System Architecture Diagram

8.References

[1] J. Han, M. Kamber, and J. Pei, Data Mining: Concepts and Techniques, 3rd ed. Morgan Kaufmann, 2012.

[2] T. Hastie, R. Tibshirani, and J. Friedman, The Elements of Statistical Learning, Springer, 2009.

[3] V. Kumar and W. Reinartz, Customer Relationship Management: Concept, Strategy, and Tools, 2nd ed. Springer, 2018.

[4] A. L. Hughes, Strategic Database Marketing, McGraw-Hill, 2006.

[5] I. Witten, E. Frank, M. Hall, and C. Pal, Data Mining: Practical Machine Learning Tools and Techniques, 4th ed. Morgan Kaufmann, 2016.

[6] C. Ware, Information Visualization: Perception for Design, 4th ed. Morgan Kaufmann, 2021.

[7] S. Few, Information Dashboard Design: Displaying Data for At-a-Glance Monitoring, Analytics Press, 2013.

[8] P. Kotler and K. L. Keller, Marketing Management, 15th ed. Pearson, 2016.

[9] F. Provost and T. Fawcett, Data Science for Business, O'Reilly Media, 2013.

[10] G. Shmueli, N. Patel, and P. Bruce, Data Mining for Business Analytics, Wiley, 2017.

[11] K. P. Murphy, Machine Learning: A Probabilistic Perspective. Cambridge, MA, USA: MIT Press, 2012.

[12] A. Géron, Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2019.

[13] C. M. Bishop, Pattern Recognition and Machine Learning. New York, NY, USA: Springer, 2006.

[14] T. Davenport and J. Harris, Competing on Analytics: The New Science of Winning. Boston, MA, USA: Harvard Business School Press, 2007.

[15] S. Russell and P. Norvig, Artificial Intelligence: A Modern Approach, 4th ed. Hoboken, NJ, USA: Pearson, 2021.